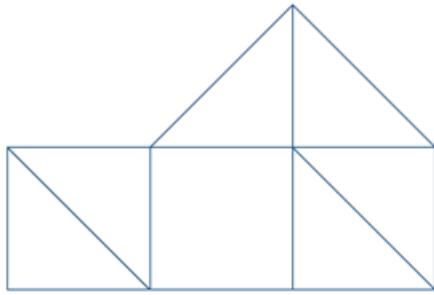


Problem

For the network graph in Fig. 2.69, find the number of nodes, branches, and loops.

Figure 2.69:



Step-by-step solution

Step 1 of 3

Refer to the network graph in Figure 2.69 in the textbook.

Redraw the network graph with node identification.

A node is the point of connection between two or more branches.

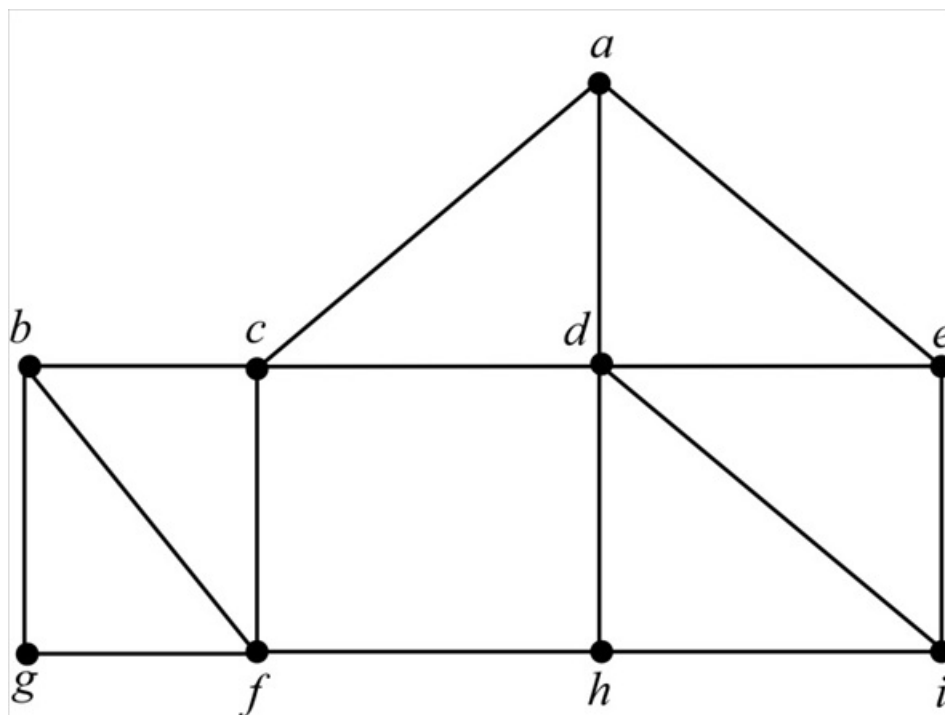


Figure 1

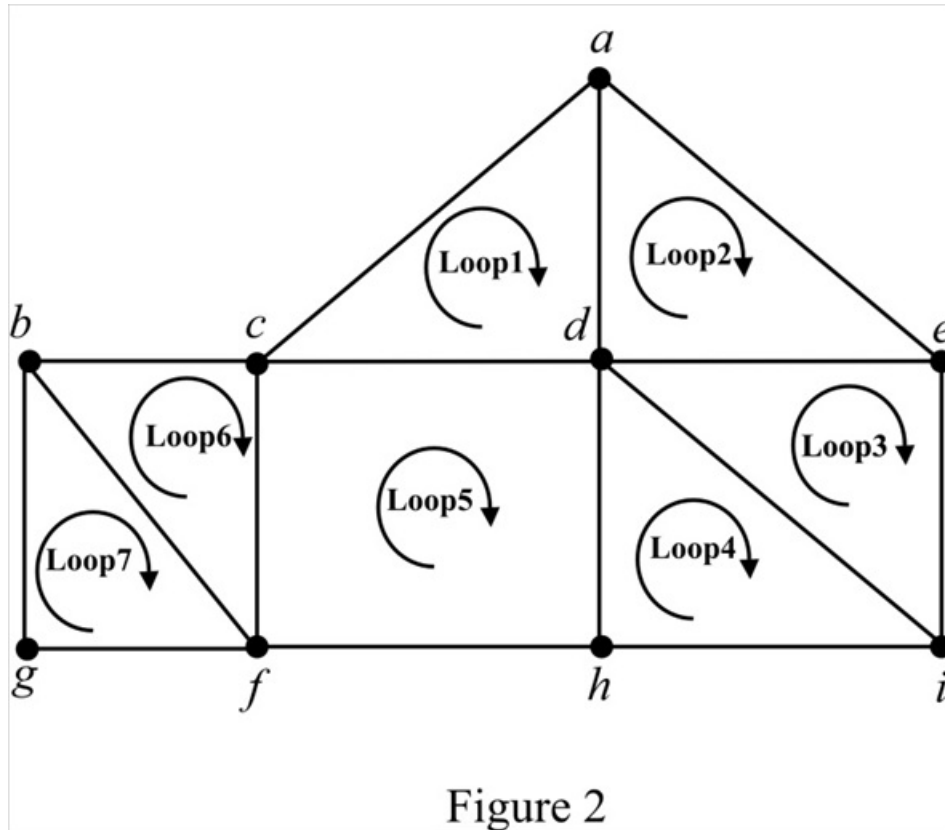
From Figure 1, there are nine nodes from (a to i).

So the numbers of nodes are .

Step 2 of 3

Redraw the network graph branch identification.

A loop is any closed path in a circuit.



From Figure 2, circuit has seven (loop 1 to loop 7).

Therefore the number of loop l in the circuit is 7.

Step 3 of 3

The fundamental theorem of network topology is,

$$\begin{aligned} b &= l + n - 1 \\ &= 7 + 9 - 1 \\ &= 16 - 1 \\ &= 15 \end{aligned}$$

Therefore number of branches b in the circuit is 15.